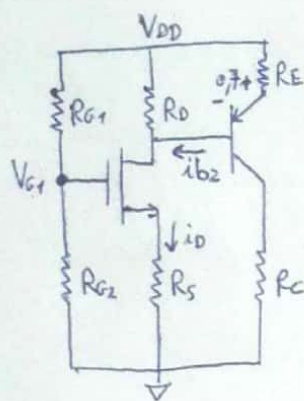


QUARTO APPELLO

1)

DC)



$$V_{G1} = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} = 4,5V$$

$$V_{GS} = V_{G1} - V_S = V_{G1} - i_D R_S = V_{G1} - R_S i_{DSS} \left(1 - \frac{V_{GS}}{V_t}\right)^2 =$$

$$= V_{G1} - R_S i_{DSS} \left(\frac{V_{GS}^2}{V_t^2} - 2 \frac{V_{GS}}{V_t} + 1\right)$$

$$\Rightarrow \left(\frac{R_S i_{DSS}}{V_t^2}\right) V_{GS}^2 + \left(1 - 2 \frac{R_S i_{DSS}}{V_t}\right) V_{GS} - (V_{G1} + R_S i_{DSS}) = 0$$

$$V_{GS} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \begin{cases} 4,16V & \text{OK} \\ -2,16V & \text{NOT OK} \end{cases}$$

Because if an nMOS is in saturation then V_{GS} must be greater than 0.

$$V_S = V_G - V_{GS} = 0,34V$$

$$i_D = i_{DSS} \left(1 - \frac{V_{GS}}{V_t}\right)^2 = 336 \mu A$$

$$(i_D - i_{b2}) R_D = (\beta_F + 1) i_{b2} R_E + 0,7 \Rightarrow i_{b2} = \frac{R_D i_D - 0,7}{R_D + (\beta_F + 1) R_E} = 5,18 \mu A$$

$$V_E = V_{DD} - (\beta_F + 1) i_{b2} R_E = 7,85V$$

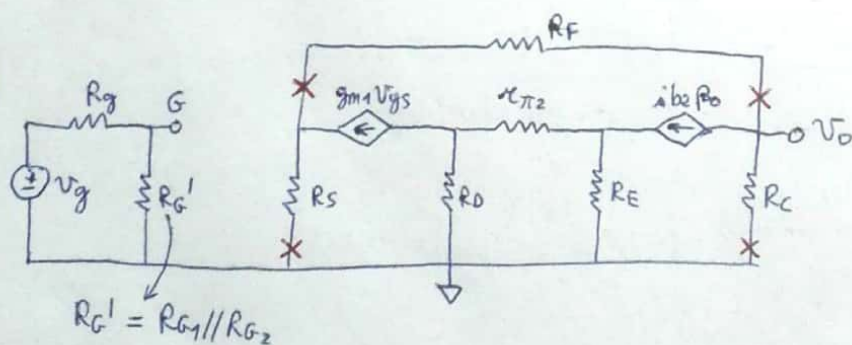
$$V_C = \beta_F i_{b2} R_C = 5,18V$$

$$\Rightarrow V_{EC} = 2,67V$$

$$V_D = V_{DD} - (i_D - i_{b2}) R_D = 7,15V$$

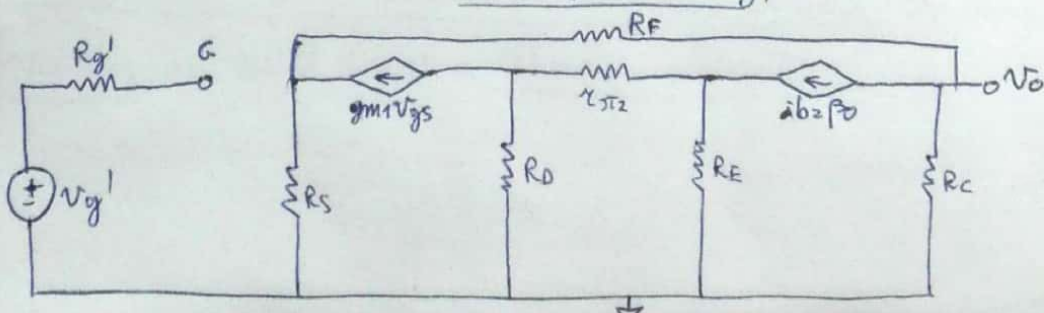
$$\Rightarrow V_{DS} = 6,81V$$

AC)



IT'S A VOLTAGE MIXING
VOLTAGE SAMPLING

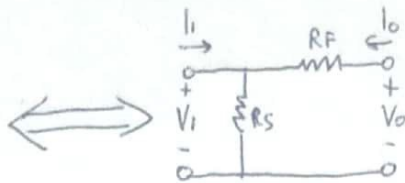
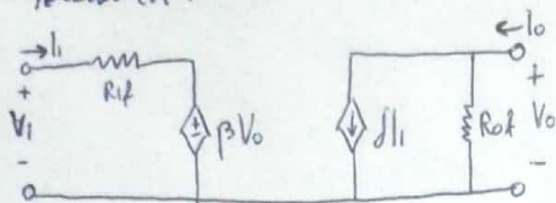
Be careful! It's a common mistake to leave the input circuit drawn as I did.
The input circuit must always be simplified with a Norton or Thevenin circuit.
In this case we have a voltage mixing, so we'll draw a Thevenin circuit.



$$v_g' = v_g \frac{R_g'}{R_g' + R_g}$$

$$R_g' = R_g // R_G$$

β -network:



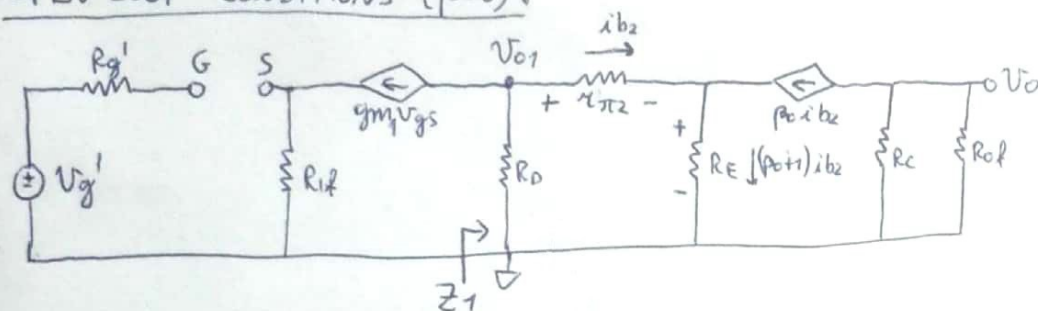
$$R_{1f} = \frac{V_1}{I_1} \Big|_{V_o=0} = R_S // R_F$$

$$R_{of} = \frac{V_o}{I_o} \Big|_{I_1=0} = R_F + R_S$$

$$\beta = \frac{V_1}{V_o} \Big|_{I_1=0} = \frac{R_S}{R_S + R_F}$$

$$\beta \rightarrow 0$$

OPEN LOOP CONDITIONS ($\beta=0$):



$$g_{m1} = \frac{2}{V_E} \sqrt{I_{D1} I_{DSS}} = 580 \mu S$$

$$g_{m2} = 40 I_{C2} = 20,72 mS$$

$$r_{\pi 2} = 4,83 k\Omega$$

$$A_d = \frac{V_o}{V_{g'}} = \frac{V_o}{V_{01}} \cdot \frac{V_{01}}{V_{g'}} = A_{v2} \cdot A_{v1}$$

$$V_o = -i_{b2} \beta_0 \cdot R_C // R_{of}$$

$$V_{01} = i_{b2} \cdot r_{\pi 2} + i_{b2} (\beta_0 + 1) R_E = i_{b2} (r_{\pi 2} + (\beta_0 + 1) R_E)$$

$$V_{01} = g_{m1} V_{gs} (R_D // (r_{\pi 2} + (\beta_0 + 1) R_E))$$

} I must find V_{01} in function of i_{b2} and V_{gs} if I want to find A_{v2} and A_{v1}

$$V_{gs} = V_{g'} - V_S = V_{g'} - g_{m1} V_{gs} R_{1f} \Rightarrow V_{g'} = (1 + g_{m1} R_{1f}) V_{gs}$$

$$A_{OL} = A_{v2} \cdot A_{v1} = \frac{-i_{b2} \cdot \beta_0 \cdot R_C // R_{of}}{i_{b2} (r_{\pi 2} + (\beta_0 + 1) R_E)} \cdot \frac{-g_{m1} V_{gs} (R_D // (r_{\pi 2} + (\beta_0 + 1) R_E))}{(1 + g_{m1} R_{1f}) V_{gs}} =$$

$$= \frac{-\beta_0 R_C // R_{of}}{r_{\pi 2} + (\beta_0 + 1) R_E} \cdot \frac{-g_{m1} (R_D // (r_{\pi 2} + (\beta_0 + 1) R_E))}{1 + g_{m1} R_{1f}} = (-2,98) \cdot (-2,04) = 6,08$$

CLOSED LOOP CONDITIONS:

$$A_v = A_F = \frac{A_d}{1 + \beta A_d} = 4,72$$

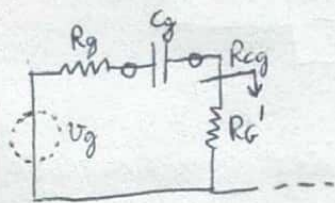
$$R_{in}^{OL} = R_{G'} = 50 k\Omega$$

NOTE: R_{in} must be calculated in the input circuit drawn without the Thevenin simplification

$R_{in}^F = R_{in}^{OL}$ because the feedback doesn't touch the node where $R_{G'}$ is connected (there's no interaction between R_{in}^{OL} and the feedback)

f_L
According to the test instructions we can neglect C_F

We draw the circuit where C_g is present:



We are interested only in resistances and capacitors. We don't care about voltage/current generators → VOLTAGE GEN. ARE REPLACED WITH SHORT CIRCUITS
CURRENT GEN. ARE REPLACED WITH OPEN CIRCUITS

$$R_{cg} = R_{g'} + R_g = 51 \text{ k}\Omega$$

R_{cg} is the resistance the capacitor sees

$$C_{cg} = C_g = 10 \text{ nF}$$

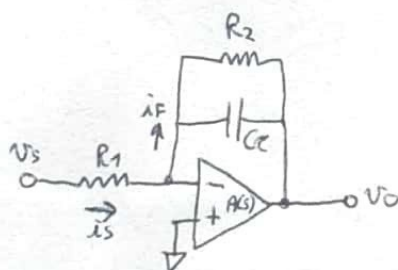
C_{cg} is the total capacitance in the circuit (in this case there's only C_g)

$$f_L^{\omega} = \frac{1}{2\pi} \frac{1}{R_{cg} C_g} = 312 \text{ Hz}$$

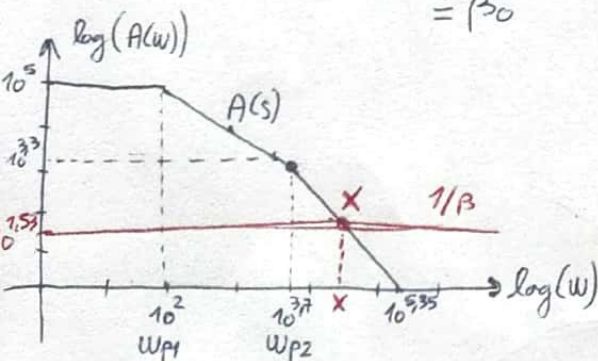
$f_L^F = f_L^{\omega}$ because the feedback doesn't interact with this capacitance

NOTE! in most of the cases $C_x = C_x$ because there's usually only one capacitance per analysed circuit

2) 1) FIND THE PM



$$\beta = \frac{V_-}{V_o} \Big|_{V_s=0} = \frac{R_1}{R_1 + R_2} = \frac{1}{34} \Rightarrow \frac{1}{\beta} = 34 = 10^{1,53}$$



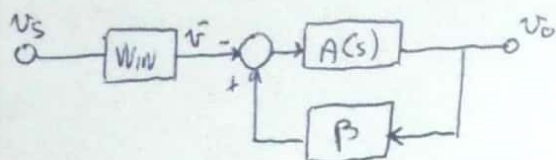
$$\frac{A_1}{A_2} = \left(\frac{w_2}{w_1} \right)^2 \Rightarrow \frac{10^{3,3}}{10^{1,53}} = \left(\frac{x}{10^{3,7}} \right)^2 \Rightarrow x = 38,46 \frac{\text{rad}}{\text{s}}$$

$$\text{PM} = 180 - \arctan\left(\frac{x}{w_{p1}}\right) - \arctan\left(\frac{x}{w_{p2}}\right) = 7,56^\circ$$

2) DRAW THE BLOCKS SCHEME OF THE CIRCUIT

According to the professor, this point of the exercise has been got wrong by most of the students. This because it's natural to draw the classic blocks scheme of the circuit which shows only voltages (as shown below on the left), but we had better to notice that there are also currents involved. In fact this amplifier is a current mixing - voltage sampling; in the mixing side there's a comparison of currents (and not of voltages).

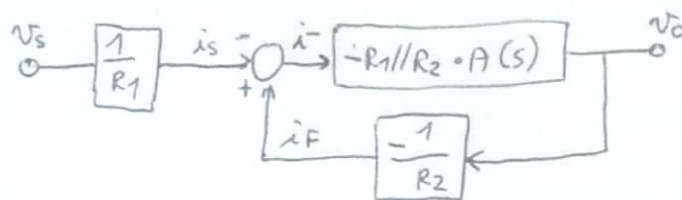
WRONG BLOCKS SCHEME!



With $W_{in} = \frac{v_s}{v_s} \Big|_{v_o=0} = \frac{Z_2}{R_1 + Z_2}$

$$\beta = \frac{R_1}{R_1 + Z_2}$$

CORRECT BLOCKS SCHEME!



$W_{in} = \frac{1}{R_1}$ IS NEEDED TO CONVERT v_s IN A CURRENT

$\beta = -\frac{1}{R_2}$ IS NEEDED TO CONVERT v_o IN A CURRENT

$-R_1//R_2$ IS THE EQUIVALENT RESISTANCE IF v_s AND v_o ARE OFF ($=0$). THE MINUS SIGN IS BECAUSE THE AMPLIFIER IS AN INVERTING ONE

3) FIND AN APPROPRIATE VALUE OF C_c TO STABILIZE THE AMPLIFIER

$$\beta = \frac{R_1}{R_1 + Z_2} = \underbrace{\frac{R_1}{R_1 + R_2}}_{\beta_0} \cdot \frac{R_2 C_c s + 1}{\frac{R_1 R_2}{R_1 + R_2} C_c s + 1} = \beta_0 \frac{R_2 C_c s + 1}{\beta_0 R_2 C_c s + 1}$$

NOTE! NOW WE'RE BACK TO VOLTAGES, SO $\beta = \frac{R_1}{R_1 + Z_2}$ AS USUAL

$$Z_\beta = \frac{1}{R_2 C_c}$$

$$P_\beta = \frac{1}{\beta_0 R_2 C_c} = \frac{1}{\beta_0} \frac{1}{R_2 C_c} = 34 Z_\beta$$

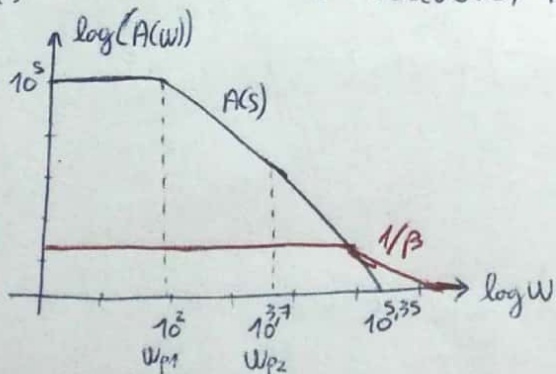
We have two solutions $\left\{ \begin{array}{l} Z_\beta = 10^2 \text{ IF WE WANT TO STABILIZE JUST THE FLATBAND} \\ Z_\beta = x = 38,46 \frac{\text{K}^{\text{rad}}}{\text{s}} \text{ IF WE WANT TO STABILIZE ALL THE BAND} \end{array} \right.$

↑ THE INTERSECTION POINT FOUND BEFORE

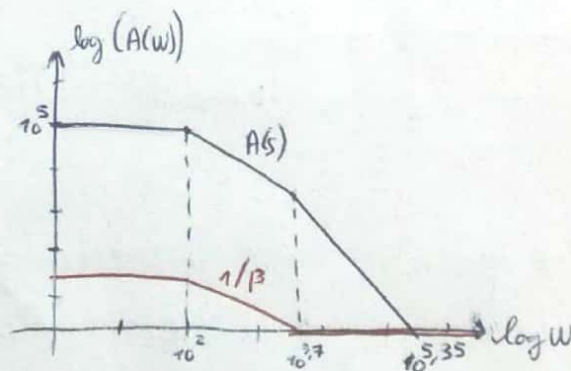
I) $Z_\beta = x \Rightarrow C_c = 788 \text{ pF}$

II) $Z_\beta = 10^2 \Rightarrow C_c = 303 \text{ nF}$

4) DRAW THE NEW FREQUENCY RESPONSE



SOLUTION 1



SOLUTION 2